For weeks 1-6, I didn’t take any notes since all the linear algebra concepts are not new to me. The only thing that was different was just the new notation, which is not hard to learn.

MC1 Practice Problems: Quantum Foundations

CS 480A8 · Spring 2026

Notation Reference

Basis States: $|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$, $|1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

Pauli Gates: $\mathbf{X} = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$, $\mathbf{Y} = \begin{pmatrix}0 & -i\\i & 0\end{pmatrix}$, $\mathbf{Z} = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$

Other Gates: $\mathbf{H} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$, $\mathbf{S} = \begin{pmatrix}1 & 0\\0 & i\end{pmatrix}$, $\mathbf{T} = \begin{pmatrix}1 & 0\\0 & e^{i\pi/4}\end{pmatrix}$

Inner Product: $\langle\psi|\phi\rangle = \alpha^*\gamma + \beta^*\delta$ for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$

Measurement: For state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$: $P(0) = |\alpha|^2$, $P(1) = |\beta|^2$

Normalization: $|\alpha|^2 + |\beta|^2 = 1$ **Global Phase:** $|\psi\rangle \equiv e^{i\theta}|\psi\rangle$ (physically equivalent)

Adjoint: $A^\dagger = (A^*)^T$ (conjugate transpose) **Unitary:** $U^\dagger U = \mathbf{I}$

1 Linear Algebra Review

$X = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$ $Y = \begin{pmatrix}0 & -i\\i & 0\end{pmatrix}$ $Z = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$

$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$ $S = \begin{pmatrix}1 & 0\\0 & i\end{pmatrix}$ $T = \begin{pmatrix}1 & 0\\0 & e^{i\pi/4}\end{pmatrix}$

Easy

0.1. [Easy] Compute the following for complex numbers:

(a) The magnitude $|3 + 4i|$ 25

(b) The complex conjugate $(2 - i)^*$ $2+i$

(c) The product $(1 + i)(1 - i)$ 2

0.2. [Easy] Compute the matrix-vector product:

$$\begin{pmatrix}2 & 1\\0 & 3\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}4\\6\end{pmatrix}$$

0.3. [Easy] Compute the transpose A^T of the matrix:

$$A = \begin{pmatrix}1 & 2\\3 & 4\end{pmatrix} \begin{pmatrix}1 & 3\\2 & 4\end{pmatrix}$$

Medium

0.4. [Medium] Compute the adjoint (conjugate transpose) $A^\dagger$ of the matrix:

$$A = \begin{pmatrix}1 & i\\2 - i & 3\end{pmatrix} \begin{pmatrix}1 & 2+i\\-i & 3\end{pmatrix}$$

0.5. [Medium] Let $\mathbf{u} = \begin{pmatrix}1\\i\end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix}i\\1\end{pmatrix}$. Compute the inner product $\mathbf{u}^\dagger\mathbf{v}$. $\begin{pmatrix}-i & 1\end{pmatrix}\begin{pmatrix}i\\1\end{pmatrix} = 2$

0.6. [Medium] Let $A = \begin{pmatrix}1 & 2\\0 & 1\end{pmatrix}$, $B = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$, and $\mathbf{v} = \begin{pmatrix}1\\1\end{pmatrix}$. Verify that $(AB)\mathbf{v} = A(B\mathbf{v})$.

$AB = \begin{pmatrix}2 & 1\\1 & 0\end{pmatrix}$ $\begin{pmatrix}3\\1\end{pmatrix}$

$\begin{pmatrix}1 & 2\\0 & 1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} \to \begin{pmatrix}3\\1\end{pmatrix}$

Hard

0.7. [Hard] A matrix U is *unitary* if $U^\dagger U = \mathbf{I}$. Determine whether the following matrix is unitary:

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \quad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \to \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark$$

0.8. [Hard] Prove that unitary matrices preserve inner products. That is, for any unitary matrix U and vectors $\mathbf{u}, \mathbf{v}$, show that:

$$(U\mathbf{u})^\dagger (U\mathbf{v}) = \mathbf{u}^\dagger \mathbf{v}$$

$u^\dagger I v$

$(u^\dagger \cancel{U^\dagger})(\cancel{U} v) = u^\dagger v$

Hint: Use the property $(AB)^\dagger = B^\dagger A^\dagger$ and the definition of unitary.

2 Dirac Notation

Easy

1.1. [Easy] Write the state $|+\rangle$ as a column vector. $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

1.2. [Easy] Compute the inner product $\langle 0|1\rangle$. 0

1.3. [Easy] Compute the inner product $\langle +|+\rangle$. 1

Medium

1.4. [Medium] Let $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$. Compute $\langle \psi|\psi\rangle$. 0

1.5. [Medium] Compute the inner product $\langle +|-\rangle$. What does this result tell you about these two states?

$\frac{1}{2} - \frac{1}{2} = 0$ This tells us $|+\rangle$ & $|-\rangle$ are orthogonal.

1.6. [Medium] Compute the outer product $|+\rangle\langle -|$ and express it as a 2×2 matrix.

$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \quad \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

Hard

1.7. [Hard] Show that the set $\{|+\rangle, |-\rangle\}$ forms an orthonormal basis. That is, prove:

(a) $\langle +|+\rangle = \langle -|-\rangle = 1$ (normalization) As shown above, $\frac{1}{2} + \frac{1}{2} = 1$

(b) $\langle +|-\rangle = \langle -|+\rangle = 0$ (orthogonality) As shown above $\frac{1}{2} - \frac{1}{2} = 0$

(c) $|+\rangle\langle +| + |-\rangle\langle -| = \mathbf{I}$ (completeness) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \to \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$

1.8. [Hard] A student claims: "Since $\langle 0|0\rangle = 1$ and $\langle 1|1\rangle = 1$, adding them gives 2, which violates normalization."

(a) Explain precisely why this reasoning is flawed. Why are you adding inner products both that won't tell you anything ab if they're orthogonal

(b) What would be a correct statement involving these inner products?

$\langle 0|1\rangle = 0$, which means they're orthogonal

3 State Representation

Easy

2.1. [Easy] A qubit is in state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. Verify that this state is normalized. Yes, because their magnitudes add up to 1

2.2. [Easy] Write the state $|-\rangle$ in the computational basis $\{|0\rangle, |1\rangle\}$. $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

2.3. [Easy] Convert the column vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to Dirac notation.

$|1\rangle$

Medium

2.4. [Medium] A qubit is in state $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$.

(a) Verify that this state is normalized. $\sqrt{\frac{1}{4} + \frac{3}{4}} = 1$ ✓

(b) Write $|\psi\rangle$ as a column vector. $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$

2.5. [Medium] Determine the value of α (where α is real and positive) such that the state $|\psi\rangle = \alpha|0\rangle + \frac{1}{2}|1\rangle$ is normalized. $\frac{\sqrt{3}}{2}$

2.6. [Medium] Are the states $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\phi\rangle = \frac{1}{\sqrt{2}}(-|0\rangle - |1\rangle)$ physically distinguishable? Explain your reasoning.

No, because $|\phi\rangle$ could have some $e^{i\theta}$ that makes them the same

Hard

2.7. [Hard] A student claims that $|\psi\rangle = |0\rangle + |1\rangle$ is a valid quantum state representing an equal superposition.

(a) Identify the error in this claim. Total magnitude = 2. You could just represent $|\psi\rangle$ as $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

(b) Write the correct normalized equal superposition state.

(c) If we mistakenly used the unnormalized state to calculate measurement probabilities, what would we get? How does this differ from the correct answer? 1 for each. This is a problem since the total magnitude is 2

2.8. [Hard] A valid quantum gate must be a *unitary* matrix, satisfying $U^\dagger U = \mathbf{I}$.

(a) Is the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ a valid quantum gate? Prove your answer. No. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

(b) Verify that the Hadamard gate $\mathbf{H}$ is unitary. $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\frac{1}{2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ ✓

4 Single-Qubit Gates

Easy

3.1. [Easy] Compute $\mathbf{X}|0\rangle$. $|1\rangle$

3.2. [Easy] Compute $\mathbf{H}|0\rangle$. $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$

3.3. [Easy] Compute $\mathbf{Z}|+\rangle$. Express your answer in the computational basis.

$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$

Medium

$T = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$

3.4. [Medium] Apply the Hadamard gate twice to $|0\rangle$. That is, compute $\mathbf{HH}\,|0\rangle$. What do you observe?

It goes from $|+\rangle$ back to $|0\rangle$.

3.5. [Medium] Let $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$. Compute $\mathbf{X}\,|\psi\rangle$ and simplify your answer.

$\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \to |+\rangle$

3.6. [Medium] Compute $\mathbf{S}\,|+\rangle$. Express your answer in the computational basis.

$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

3.7. [Medium] A qubit starts in state $|0\rangle$. We apply $\mathbf{H}$, then $\mathbf{T}$, then $\mathbf{H}$. What is the final state?

$T|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}\ \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1+i}{\sqrt{2}} \end{pmatrix}\ \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{1}{\sqrt{2}}+\frac{1+i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}}-\frac{1-i}{\sqrt{2}} \end{pmatrix}$

3.8. [Medium] Compute $\mathbf{H}\,|-\rangle$ and simplify.

$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$

(a) Write out the calculation showing how the amplitudes of $|0\rangle$ and $|1\rangle$ combine.

(b) Identify which amplitude undergoes constructive interference and which undergoes destructive interference.

(c) What is the final state?

Hard

3.9. [Hard] Prove that $\mathbf{HXH} = \mathbf{Z}$ by:

$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(a) Computing the matrix product directly, OR

(b) Showing that both sides act identically on the basis states $|0\rangle$ and $|1\rangle$

3.10. [Hard] Consider the gate sequence $\mathbf{X} \to \mathbf{H} \to \mathbf{Z} \to \mathbf{H}$ applied to an arbitrary state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$.

(a) Write down the combined unitary matrix for this sequence.

$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ $\begin{pmatrix} \beta \\ \alpha \end{pmatrix}$ $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} \beta+\alpha \\ \alpha-\beta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

(b) Compute the final state.

$\begin{pmatrix} \beta+\alpha \\ \alpha-\beta \end{pmatrix}\begin{pmatrix} \beta+\alpha \\ \beta-\alpha \end{pmatrix}$

(c) What single gate is this sequence equivalent to?

(Remember: gates are applied right-to-left in matrix notation.)

5 Measurement

Easy

4.1. [Easy] A qubit is in state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. What is the probability of measuring 0 in the computational basis?

$\frac{1}{2}$

4.2. [Easy] A qubit is measured in the computational basis and the outcome is 1. What is the post-measurement state?

4.3. [Easy] A qubit is in state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. What are the probabilities of measuring 0 and 1?

$P(0) = \frac{9}{25}$, $P(1) = \frac{16}{25}$

Medium

4.4. [Medium] A qubit is in state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$.

(a) What is $P(\text{measuring } 0)$? $\frac{1}{3}$

(b) What is $P(\text{measuring } 1)$? $\frac{2}{3}$

(c) Verify that these probabilities sum to 1. bulk

4.5. [Medium] A qubit is prepared in state $|0\rangle$, then a Hadamard gate is applied, and finally the qubit is measured in the computational basis.

(a) What is the state immediately before measurement? $|+\rangle$

(b) What are the possible outcomes and their probabilities? $P(0)=\frac{1}{2}$, $P(1)=\frac{1}{2}$

(c) If the measurement outcome is 0, what is the post-measurement state?

4.6. [Medium] A qubit is in state $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}e^{i\pi/4}|1\rangle$. Calculate the probability of measuring 1. Does the complex phase affect the measurement probability? $\frac{3}{4}$. No.

4.7. [Medium] The expectation value of an observable $\mathbf{Z}$ for a state $|\psi\rangle$ is $\langle \mathbf{Z}\rangle = \langle\psi|\,\mathbf{Z}\,|\psi\rangle$.

(a) Compute $\langle \mathbf{Z}\rangle$ for the state $|0\rangle$. $\langle 0|Z|0\rangle$

(b) Compute $\langle \mathbf{Z}\rangle$ for the state $|+\rangle$.

(c) Interpret your results: what does the expectation value tell you about the average of many measurements?

4.8. [Medium] A qubit is in state $|+\rangle$.

(a) You measure in the computational basis and get outcome 0. What is the post-measurement state?

(b) You immediately apply $\mathbf{H}$ to this post-measurement state. What is the resulting state?

(c) You measure again in the computational basis. What are the probabilities of 0 and 1?

Hard

4.9. [Hard] A qubit is in an unknown state $|\psi\rangle$. You apply the sequence: $\mathbf{X} \to \mathbf{H} \to$ measure. The measurement yields 0 with probability 1 (i.e., you always get 0). Determine the original state $|\psi\rangle$.

Hint: Work backwards from the measurement constraint.

4.10. [Hard] Consider measuring in the $\{|+\rangle, |-\rangle\}$ basis instead of the computational basis. For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

(a) Express $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis. That is, find c_+ and c_- such that $|\psi\rangle = c_+|+\rangle + c_-|-\rangle$.

(b) What is the probability of obtaining $|+\rangle$ when measuring in this basis?

(c) A qubit is prepared in state $|0\rangle$. If we measure in the $\{|+\rangle, |-\rangle\}$ basis, what are the probabilities of each outcome?

Tensor products & entanglement

Tensor product $|0\rangle \otimes |1\rangle$ or $|01\rangle$

- combines 2 qubit spaces - $\mathbb{C}^2 \otimes \mathbb{C} = \mathbb{C}^4$
- Distribute & multiply: $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$
- properties
  - bilinear: $(|\psi_1\rangle + |\psi_2\rangle) \otimes |\phi\rangle = |\psi_1\phi\rangle + |\psi_2\phi\rangle$
  - Kronecker product $\longrightarrow \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a\begin{pmatrix} c \\ d \end{pmatrix} \\ b\begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}$
  - not commutative, i.e. $|01\rangle \neq |10\rangle$
  - associative, i.e. $(|a\rangle \otimes |b\rangle) \otimes |c\rangle = |a\rangle \otimes (|b\rangle \otimes |c\rangle) = |abc\rangle$
  - dimension: $\dim(V \otimes W) = \dim(V)\dim(W)$
  - normalization: $\||\psi\rangle\| = 1$ && $\||\phi\rangle\| = 1 \Rightarrow \||\psi\rangle \otimes |\phi\rangle\| = 1$

Exponential scaling

- each qubit doubles state space, i.e. $|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$
- we can manipulate an exponential amount of amplitudes with a linear amount of qubits. we can only see n things, but theoretically we can manipulate 2^n bits.

is this state separable?

given $|\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$, reduce to $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$

$\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta = \frac{1}{2}$ $\quad \alpha = \frac{1}{\sqrt{2}}, \beta = \frac{1}{\sqrt{2}}, \gamma = \frac{1}{\sqrt{2}}, \delta = \frac{1}{\sqrt{2}}$

$|\psi\rangle = |+\rangle \otimes |+\rangle$

$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$\alpha\gamma = \frac{1}{\sqrt{2}}$ $\quad \alpha\delta = 0$ $\quad$ if qubit a = 0, qubit b = 0

$\beta\gamma = 0$ $\quad \beta\delta = \frac{1}{\sqrt{2}}$ $\quad$ if qubit a = 1, qubit b = 1

} correlated → entanglement

$\left.\begin{matrix} \alpha \neq 0, \gamma \neq 0 \\ \beta \neq 0, \delta \neq 0 \end{matrix}\right]$ - contradiction

$|0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0\begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0\begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $\quad \frac{1}{\sqrt{2}}\begin{pmatrix} 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & 0\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ 0\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{pmatrix}$

$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} |00\rangle$

Interference & the power of quantum computing

Gates in multi-qubit systems

- to apply H to a 2-qubit system (qubit 2), compute $I \otimes H$, giving you a 4×4
- $(A \otimes B)(|a\rangle \otimes |b\rangle) = A|a\rangle \otimes B|b\rangle$

$$(I \otimes H)|00\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = I|0\rangle \otimes H|0\rangle = |0\rangle \otimes |+\rangle$$

- CNOT cannot be factored into a pre-tensor state, making it a truly multiqubit gate
- $CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$, $CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$, $SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$|+\rangle \to |1\rangle$ $\quad$ $|10\rangle \to |11\rangle$, $|11\rangle \to |10\rangle$ $\quad$ $|ab\rangle \to |ba\rangle$

$|00\rangle \to H \otimes I \to CNOT$

$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$ $\quad$ $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

$SWAP = CNOT_{12} \circ CNOT_{21} \circ CNOT_{12}$

$$|01\rangle \xrightarrow[CNOT_{12}]{} |01\rangle \xrightarrow[CNOT_{21}]{} |11\rangle \xrightarrow[CNOT_{12}]{} |10\rangle$$

- Any complicated gate is a combo of 1-qubit gates & CNOT

Superposition

- Non-trivial linear combination of basis states – $|\psi\rangle = \sum \alpha_x |x\rangle$ where more than $\alpha_x \neq 0$
- entanglement is multiqubit superposition that can't be factored into individual qubit states
- Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ – measuring qubit a instantly determines qubit b
- All entangled states are in superposition, but not all states in superposition is entangled
  - Separability: if n qubits are separable, each described independently with $2n$ params – scales linearly
  - If entangled, you must track every amplitude – scales exponentially

Interference

- amplitudes can interfere constructively or destructively unlike in classical

$|0\rangle \xrightarrow[H]{} |+\rangle \xrightarrow[H]{} |0\rangle$ $\quad$ $|-\rangle$ destructively interferes with $|1\rangle$

Engineering entanglement

Recall $H|0\rangle \to |+\rangle \xrightarrow[H]{} |0\rangle$

Given a superposition of equal amplitudes, concentrate all probability to $|11\rangle$

Apply $H \otimes H$ to $|00\rangle \to \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$

Apply CZ, which only flips $|11\rangle$'s sign $\to \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$

Apply $H \otimes H$, then phase flip, then $H \otimes H$ again

Example: find 1 operation to get only $|11\rangle$ from $CZ|+\rangle$

$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ $\qquad (I - CZ)|\psi\rangle = |11\rangle$

$CZ|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ $\qquad$ but $(I - CZ)$ isn't unitary

How to do $(I - CZ)$ in a quantum computer?

- embed the operation in a larger unitary circuit with an ancilla (helper qubit)
- ancilla creates 2 paths through circuit; interference between them produces desired linear combo

Example: ancilla $= |0\rangle$, $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

Apply H to ancilla $\to |\psi\rangle \frac{1}{\sqrt{2}}(|0,00\rangle + |0,01\rangle + |0,10\rangle + |0,11\rangle + |1,00\rangle + |1,01\rangle + |1,10\rangle + |1,11\rangle)$

Apply CZ only when ancilla is $|1\rangle \to \frac{1}{\sqrt{2}}|0\rangle|\psi\rangle + \frac{1}{\sqrt{2}}|1\rangle(CZ|\psi\rangle)$

Apply H again $\to \frac{1}{2}|0\rangle(|00\rangle + |01\rangle + |10\rangle) + \frac{1}{2}|1\rangle(I - CZ)|\psi\rangle = \frac{1}{2}|0\rangle(|00\rangle + |01\rangle + |10\rangle) + \frac{1}{2}|1\rangle|11\rangle$

Measure where ancilla is $|1\rangle$ to get the result

Teleportation & no-cloning

No-cloning theorem

- No unitary operation can copy an arbitrary unknown state – $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$
- Proof

Assume U exists such that $U|a\rangle\otimes|0\rangle = |a\rangle\otimes|a\rangle$ & $U|b\rangle\otimes|0\rangle = |b\rangle\otimes|b\rangle$

$\langle aa|bb\rangle = \langle a|b\rangle\cdot\langle a|b\rangle$: $z = \langle a|b\rangle \rightarrow z = z^2$, implying $z(z-1)=0 \rightarrow z\in\{0,1\}$

$\therefore$ There is no U that allows for a or b to be anything but identical or orthogonal

- Implications
  - can't backup quantum state, but can teleport them
  - can't eavesdrop perfectly – quantum key distribution is possible
  - can measure to destroy information

Teleportation

A has $|\psi\rangle$ wants to send to Bob

- 3 qubits
  - q_0: A's unknown state $\alpha|0\rangle+\beta|1\rangle$
  - q_1: A's Bell pair
  - q_2: B's Bell pair

1. $CNOT(q_0 \rightarrow q_1)$, then H_{q_0}
2. Measure q_0 & q_1, getting 1 of 4 outcomes
   - $\frac{1}{2}|00\rangle|\psi\rangle$, – original
   - $\frac{1}{2}|01\rangle|\psi\rangle$ – apply X
   - $\frac{1}{2}|10\rangle|\psi\rangle$ – apply Z
   - $\frac{1}{2}|11\rangle|\psi\rangle$ – apply ZX

MC2 review: multi-qubit, Bell states, local operations, tensor products, multi-qubit gates

Born rule

$|\psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle \quad - \quad P(x) = |c_x|^2$ $\qquad |\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$

e.g. $|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ $\qquad P(00) = \frac{1}{4}, P(01) = \frac{1}{4}, P(10) = \frac{1}{2}, P(11) = 0$

$P(00) = \frac{1}{2}, \; P(11) = \frac{1}{2}$

$P(01) = 0, \; P(10) = 0$

Partial measurement - just only care about $P(q_1 = x)$

$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$

$P(q_1 = 0) = |\alpha_{00}|^2 + |\alpha_{01}|^2$

Post-measurement: $\frac{\alpha_{00}|00\rangle + \alpha_{01}|01\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{01}|^2}}$

Measurement non-uniqueness

- Not every state is unique

Bell states

$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

$|\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Local operations

CNOT creates entangled states $\qquad |00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Tensor operations

$|+\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle$

$|\psi\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle + 0 \quad (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$

$\alpha\gamma = \frac{1}{2}, \; \alpha\delta = -\frac{1}{2}, \; \beta\gamma = \frac{1}{2}, \; \beta\delta = -\frac{1}{2}$

$\alpha = \frac{1}{\sqrt{2}}, \; \beta = \frac{1}{\sqrt{2}}, \; \gamma = \frac{1}{\sqrt{2}}, \; \delta = -\frac{1}{\sqrt{2}}$

$\langle\alpha|\alpha\rangle = 1, \; \langle\beta|\beta\rangle = 1$

Prove $|\alpha\rangle \otimes |\beta\rangle$ also has a magnitude of 1.

MCQ review cont'd.

$|\psi\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$ factor into prod

$\alpha\gamma = \frac{1}{2}, \alpha\delta = -\frac{1}{2}, \beta\gamma = \frac{1}{2}, \beta\delta = -\frac{1}{2}$ $\quad \alpha = \frac{1}{\sqrt{2}}\ \beta = \frac{1}{\sqrt{2}}\ \gamma = \frac{1}{\sqrt{2}}\ \delta = -\frac{1}{\sqrt{2}}$

$|+\rangle \otimes |-\rangle$

$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{i}{\sqrt{2}}|11\rangle$

$\alpha\gamma = \frac{1}{\sqrt{2}}, \alpha\delta = 0, \beta\gamma = 0, \beta\delta = \frac{-i}{\sqrt{2}}$

Contradiction: δ can't be 0 and nonzero complex $\rightarrow$ entangled

When is $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{e^{i\theta}}{\sqrt{2}}|11\rangle$

$\alpha\gamma = \frac{1}{\sqrt{2}}, \beta\delta = \frac{e^{i\theta}}{\sqrt{2}}$

$\alpha\delta$ must be $0 \rightarrow \delta$ must be 0, but that can't be true bc $\beta\delta \neq 0$

Entangled for all θ

Apply $H \otimes H$ to $|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$

$\psi = |-\rangle \otimes |-\rangle \quad (H \otimes H)(|-\rangle \otimes |-\rangle) \rightarrow |11\rangle$

Apply CZ to $|++\rangle$

$|++\rangle = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}$ CZ flips the sign of $|11\rangle \rightarrow \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle$

5.5 ancilla: $|0\rangle$, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$H \downarrow CZ$

$|+\rangle$

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \otimes \begin{pmatrix}\alpha\\\beta\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha\\\beta\\\alpha\\\beta\end{pmatrix}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&-1\end{pmatrix} \rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}\alpha\\\beta\\\alpha\\-\beta\end{pmatrix}$$

$$\frac{1}{\sqrt{2}}(\alpha|00\rangle + \beta|01\rangle + \alpha|10\rangle - \beta|11\rangle)$$

7.7

$|\psi\rangle = |+\rangle, \beta_{00} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad \alpha, \beta = \frac{1}{\sqrt{2}} \xrightarrow{CNOT} \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|110\rangle + \frac{\beta}{\sqrt{2}}\sqrt{101}$

$|\psi\rangle_0 \otimes |\Phi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

$H_{p0} \downarrow$

$\frac{1}{2}|00\rangle(\alpha|0\rangle + \beta|1\rangle) + \frac{1}{2}|01\rangle(\beta|0\rangle + \alpha|1\rangle)$

$+ \frac{1}{2}|10\rangle(\alpha|0\rangle - \beta|1\rangle) + \frac{1}{2}|11\rangle(-\beta|0\rangle + \alpha|1\rangle)$

A gets $|11\rangle \rightarrow$ B needs to apply ZX. His state before is $-\beta|0\rangle + \alpha|1\rangle$

Post-measurement state: $|11\rangle$